

MATHEMATICS

TARGET : JEE- Advanced

CAPS-20

VECOTR

SCQ (Single Correct Type) :

- The vector $\hat{i} + x\hat{j} + 3\hat{k}$ is rotated through an angle of $\cos^{-1} \frac{11}{14}$ and doubled in magnitude, then it becomes $4\hat{i} + (4x-2)\hat{j} + 2\hat{k}$. The value of 'x' is:
 (A) $-\frac{2}{3}$ (B) $\frac{2}{3}$ (C) $\frac{1}{3}$ (D) 2
- If the acute angle that the vector, $\alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}$ makes with the plane of the two vectors $2\hat{i} + 3\hat{j} - \hat{k}$ and $\hat{i} - \hat{j} + 2\hat{k}$ is $\cot^{-1} \sqrt{2}$ then :
 (A) $\alpha(\beta + \gamma) = \beta\gamma$ (B) $\beta(\gamma + \alpha) = \gamma\alpha$
 (C) $\gamma(\alpha + \beta) = \alpha\beta$ (D) $\alpha\beta + \beta\gamma + \gamma\alpha = 0$
- If $\vec{p}, \vec{q}, \vec{r}$ are three mutually perpendicular vectors of the same magnitude and if a vector $\vec{x}$ satisfies the equation $\vec{p} \times [(\vec{x} - \vec{q}) \times \vec{p}] + \vec{q} \times [(\vec{x} - \vec{r}) \times \vec{q}] + \vec{x} \times [(\vec{x} - \vec{p}) \times \vec{r}] = \vec{0}$, then vector $\vec{x}$ is equal to _____.
 (A) $\frac{1}{2}(\vec{p} + \vec{q} - 2\vec{r})$ (B) $\frac{1}{2}(\vec{p} + \vec{q} + \vec{r})$ (C) $\frac{1}{3}(\vec{p} + \vec{q} + \vec{r})$ (D) $\frac{1}{3}(2\vec{p} + \vec{q} - \vec{r})$
- Let $\vec{a}$ and $\vec{b}$ be two vectors of equal magnitude of 5 units each. Let $\vec{p}$ and $\vec{q}$ be vectors such that $\vec{p} = \vec{a} + \vec{b}$ and $\vec{q} = \vec{a} - \vec{b}$. If $|\vec{p} \times \vec{q}| = 2\left\{\lambda - (\vec{a} - \vec{b})^2\right\}^{\frac{1}{2}}$, then value of λ is _____.
 (A) 25 (B) 125 (C) 625 (D) None of these
- $|\vec{a}| = |\vec{b}| = |\vec{c}| = |\vec{a} + \vec{b}| = 1, \vec{a} \cdot \vec{c} = 0$ and $\vec{a} = \frac{\hat{i}}{\sqrt{2}} + \frac{\hat{j}}{\sqrt{2}}$, where $\vec{a}, \hat{k}$ and $\vec{b}$ are linearly dependent vectors. If for some $\vec{c}, |\vec{b} \times \vec{c}| = |\vec{a} \times \vec{c}|$ and $\vec{c} = p\hat{i} + q\hat{j} + r\hat{k}$, then $16(p^4 + q^4 + r^4)$ is _____.
 (A) 8 (B) 21 (C) $\frac{21}{3}$ (D) 4
- Let $\vec{a}, \vec{b}$ and $\vec{c}$ be three unit vectors such that $|\vec{a} + \vec{b} + \vec{c}| = \sqrt{3}$. If $(\vec{a} \times \vec{b}) \cdot (\vec{b} \times \vec{c}) + (\vec{b} \times \vec{c}) \cdot (\vec{c} \times \vec{a}) + (\vec{c} \times \vec{a}) \cdot (\vec{a} \times \vec{b}) = \lambda$, then the maximum value of λ is
 (A) 0 (B) 1 (C) $\sqrt{3}$ (D) 2
- Let $\vec{r}$ be the position vector of a variable point in the Cartesian OXY plane such that $\vec{r} \cdot (10\hat{j} - 8\hat{i} - \vec{r}) = 40$ and $P_1 = \max\left\{|\vec{r} + 2\hat{i} - 3\hat{j}|^2\right\}$, $P_2 = \min\left\{|\vec{r} + 2\hat{i} - 3\hat{j}|^2\right\}$. P_1^2 is equal to _____.
 (A) 9 (B) $2\sqrt{2} - 1$ (C) $6\sqrt{2} + 1$ (D) $9 - 4\sqrt{2}$



8. Let $\vec{r}$ be the position vector of a variable point in the Cartesian OXY plane such that $\vec{r} \cdot (10\hat{j} - 8\hat{i} - \vec{r}) = 40$ and $P_1 = \max\{|\vec{r} + 2\hat{i} - 3\hat{j}|^2\}$, $P_2 = \min\{|\vec{r} + 2\hat{i} - 3\hat{j}|^2\}$. $P_1 + P_2$ is equal to ____.
- (A) 2 (B) 10 (C) 18 (D) 5
9. Given $|\vec{a}| = |\vec{b}| = 1$ and $|\vec{a} + \vec{b}| = \sqrt{3}$. If $\vec{c}$ is a vector such that $\vec{c} - \vec{a} - 2\vec{b} = 3(\vec{a} \times \vec{b})$, then $(\vec{c} \cdot \vec{b})$ is equal to
- (A) $-\frac{1}{2}$ (B) $\frac{1}{2}$ (C) $\frac{3}{2}$ (D) $\frac{5}{2}$

MCQ (One or more than one correct) :

10. If $\vec{a} = x\hat{i} + y\hat{j} + z\hat{k}$, $\vec{b} = y\hat{i} + z\hat{j} + x\hat{k}$ and $\vec{c} = z\hat{i} + x\hat{j} + y\hat{k}$, then $\vec{a} \times (\vec{b} \times \vec{c})$ is
- (A) parallel to $(y - z)\hat{i} + (z - x)\hat{j} + (x - y)\hat{k}$ (B) orthogonal to $\hat{i} + \hat{j} + \hat{k}$
 (C) orthogonal to $(y + z)\hat{i} + (z + x)\hat{j} + (x + y)\hat{k}$ (D) orthogonal to $x\hat{i} + y\hat{j} + z\hat{k}$
11. The value(s) of $\alpha \in [0, 2\pi]$ for which vector $\vec{a} = \hat{i} + 3\hat{j} + (\sin 2\alpha)\hat{k}$ makes an obtuse angle with the z-axis and the vectors $\vec{b} = (\tan \alpha)\hat{i} - \hat{j} + 2\sqrt{\sin \frac{\alpha}{2}}\hat{k}$ and $\vec{c} = (\tan \alpha)\hat{i} + (\tan \alpha)\hat{j} - 3\sqrt{\operatorname{cosec} \frac{\alpha}{2}}\hat{k}$ are orthogonal, is/are:
- (A) $\tan^{-1} 3$ (B) $\pi - \tan^{-1} 2$ (C) $\pi + \tan^{-1} 3$ (D) $2\pi - \tan^{-1} 2$
12. Let $\vec{a}$ and $\vec{c}$ be unit vectors such that $\vec{a} \times \vec{b} = 2\vec{a} \times \vec{c}$, where $|\vec{b}| = 4$. The angle between $\vec{a}$ and $\vec{c}$ is $\cos^{-1}\left(\frac{1}{4}\right)$. If $\vec{b} - 2\vec{c} = k\vec{a}$, then k is equal to ____.
- (A) $\frac{1}{3}$ (B) $\frac{1}{4}$ (C) -4 (D) 3
13. If $(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) \cdot (\vec{a} \times \vec{d}) = 0$, then which of the following may be true?
- (A) $\vec{a}, \vec{b}, \vec{c}$ and $\vec{d}$ are necessarily coplanar (B) $\vec{a}$ lies in the plane of $\vec{c}$ and $\vec{d}$
 (C) $\vec{b}$ lies in the plane of $\vec{c}$ and $\vec{d}$ (D) $\vec{c}$ lies in the plane of $\vec{a}$ and $\vec{d}$

Numerical based Questions :

14. If $\vec{d}$ is a unit vector and $\vec{a}, \vec{b}, \vec{c}$ are three non-coplanar vectors, then the value of $\frac{(\vec{a} \cdot \vec{d})(\vec{b} \times \vec{c}) + (\vec{b} \cdot \vec{d})(\vec{c} \times \vec{a}) + (\vec{c} \cdot \vec{d})(\vec{a} \times \vec{b})}{[\vec{a} \ \vec{b} \ \vec{c}]}$ is equal to ____.
15. Let $(\hat{p} \times \vec{q}) \times \hat{p} + (\hat{p} \cdot \vec{q})\vec{q} = (x^2 + y^2)\vec{q} + (14 - 4x - 6y)\hat{p}$, where $\hat{p}$ and $\vec{q}$ are two non-collinear vectors (and $\hat{p}$ is a unit vector) and x, y are scalars. Find the value of $(x+y)$.
16. Let $\vec{r} = (\vec{a} \times \vec{b})\sin x + (\vec{b} \times \vec{c})\cos y + (\vec{c} \times \vec{a})$, where $\vec{a}, \vec{b}$ and $\vec{c}$ are non-zero non-coplanar vectors. If $\vec{r}$ is orthogonal to $3\vec{a} + 5\vec{b} + 2\vec{c}$, then the value of $\sec^2 y + \operatorname{cosec}^2 x + \sec y \operatorname{cosec} x$ is ____.

17. Given four non zero vectors $\vec{a}$, $\vec{b}$, $\vec{c}$ and $\vec{d}$. The vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ are coplanar but not collinear pair by pair and vector $\vec{d}$ is not coplanar with vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ and $(\vec{a} \wedge \vec{b}) = (\vec{b} \wedge \vec{c}) = \frac{\pi}{3}$, $(\vec{d} \wedge \vec{a}) = \alpha$ and $(\vec{d} \wedge \vec{b}) = \beta$, if $(\vec{d} \wedge \vec{c}) = \cos^{-1}(m\cos\beta + n\cos\alpha)$ then $m - n$ is :
18. Given $f^2(x) + g^2(x) + h^2(x) \leq 9$ and $U(x) = 3f(x) + 4g(x) + 10h(x)$, where $f(x)$, $g(x)$ and $h(x)$ are continuous $\forall x \in \mathbb{R}$. If maximum value of $U(x)$ is $\sqrt{N}$. Then the value of cube root of $(N-1000)$ is:
19. In an equilateral $\triangle ABC$ find the value of $\frac{|\overline{PA}|^2 + |\overline{PB}|^2 + |\overline{PC}|^2}{R^2}$ where P is any arbitrary point lying on its circumcircle, is
20. If $\vec{r}$ represents the position vector of point R in which the line AB cuts the plane CDE, where position vectors of points A, B, C, D, E are respectively $\vec{a} = \hat{i} + 2\hat{j} + \hat{k}$, $\vec{b} = 2\hat{i} + \hat{j} + 2\hat{k}$, $\vec{c} = -4\hat{j} + 4\hat{k}$, $\vec{d} = 2\hat{i} - 2\hat{j} + 2\hat{k}$ and $\vec{e} = 4\hat{i} + \hat{j} + 2\hat{k}$, then r^2 is :
21. Line L_1 is parallel to vector $\vec{\alpha} = -3\hat{i} + 2\hat{j} + 4\hat{k}$ and passes through a point A(7, 6, 2) and line L_2 is parallel to a vector $\vec{\beta} = 2\hat{i} + \hat{j} + 3\hat{k}$ and passes through a point B(5, 3, 4). Now a line L_3 parallel to a vector $\vec{r} = 2\hat{i} - 2\hat{j} - \hat{k}$ intersects the lines L_1 and L_2 at points C and D respectively, then $|4\overline{CD}|$ is equal to :

Matrix Match Type :

22. Match the following.

Column – I	Column – II
(A) If D, E and F are the mid points of the sides BC, CA and AB respectively of a triangle ABC and λ is a scalar such that $\overline{AD} + \frac{2}{3}\overline{BE} + \frac{1}{3}\overline{CF} - \lambda\overline{AC}$, then λ is equal to	(p) 0
(B) If $\vec{A}$, $\vec{B}$ and $\vec{C}$ are vectors such that $ \vec{B} = \vec{C} $, then the value of $\left(((\vec{A} + \vec{B}) \times (\vec{A} + \vec{C})) \times (\vec{B} \times \vec{C}) \right) \cdot (\vec{B} + \vec{C})$ is	(q) $\frac{1}{3}$
(C) In a $\triangle ABC$, points D, E and F are taken on the sides BC, CA and AB respectively such that $\frac{BD}{DC} = \frac{CE}{EA} = \frac{AF}{FB} = 2$. If $\text{area}(\triangle DEF) = \lambda \text{area}(\triangle ABC)$, then λ is equal to	(r) $\frac{1}{2}$
(D) The corner P of the square OPQR is folded up so that the plane OPQ is perpendicular to the plane OQR. If θ is angle between OP and QR, then $ \cos\theta $ is equal to	(s) $\frac{3}{5}$
	(t) $\frac{3}{4}$

(A) A–p; B–r; C–q; D–s

(B) A–r; B–p; C–s; D–r

(C) A–s; B–q; C–p; D–r

(D) A–r; B–p; C–q; D–r

Subjective based Questions :

23. Let ABC be a triangle. Points M, N and P are taken on the sides AB, BC and CA respectively such that $\frac{AM}{AB} = \frac{BN}{BC} = \frac{CP}{CA} = \lambda$. Prove that the vectors $\overrightarrow{AN}$, $\overrightarrow{BP}$ and $\overrightarrow{CM}$ form a triangle. Also find λ for which the area of the triangle formed by these vectors is the least.
24. Let P be an interior point of a triangle ABC and AP, BP, CP meet the sides BC, CA, AB in D, E, F respectively. Show that
- $$\frac{AP}{PD} = \frac{AF}{FB} + \frac{AE}{EC}$$
25. Let PM be the perpendicular from the point P(1, 2, 3) to the x-y plane. If OP makes an angle θ with the positive direction of the z-axis and OM makes an angle ϕ with the positive direction of the x-axis, where O is the origin, then find θ and ϕ

ANSWER KEY OF CAPS-20

- | | | | | |
|----------|----------|-----------------------------|---------|------------|
| 1. (D) | 2. (A) | 3. (B) | 4. (C) | 5. (A) |
| 6. (A) | 7. (D) | 8. (C) | 9. (D) | 10. (ABCD) |
| 11. (BD) | 12. (CD) | 13. (BCD) | 14. (1) | 15. (5) |
| 16. (3) | 17. (2) | 18. (5) | 19. (6) | 20. (18) |
| 21. (36) | 22. (D) | 23. $\lambda = \frac{1}{2}$ | | |